LIGHT – REFLECTION & REFRACTION

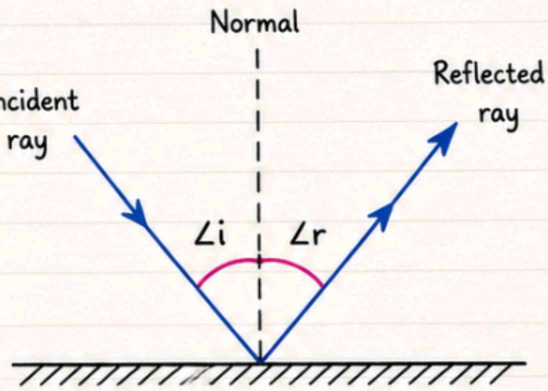
CLASS 10 SHORT NOTES

PART 1: REFLECTION OF LIGHT

Laws of Reflection (Never forget these!)

1. Angle of incidence = Angle of reflection ($\angle i = \angle r$)
2. Incident ray, normal, and reflected ray – all lie in the same plane

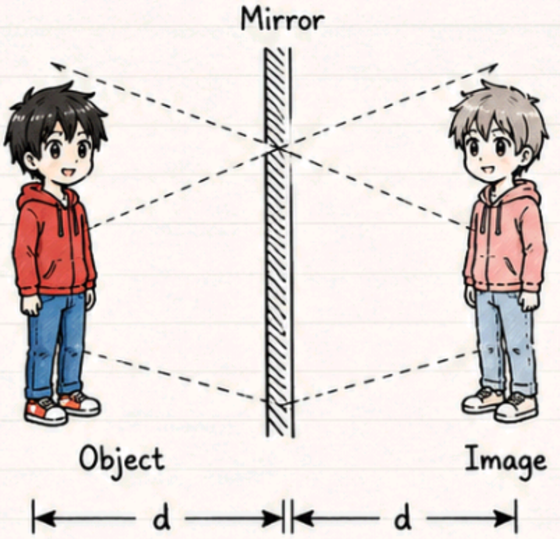
These laws apply to ALL surfaces – plane, concave, convex. Don't limit them to plane mirrors.



Plane Mirror – Image Properties

Trick to remember: "VESSEL"

- **V** Virtual
- **E** Erect
- **S** Same size as object
- **S** Same distance (image is as far behind mirror as object is in front)
- **E** Exact – Laterally inverted
- **L** Light not actually coming from image



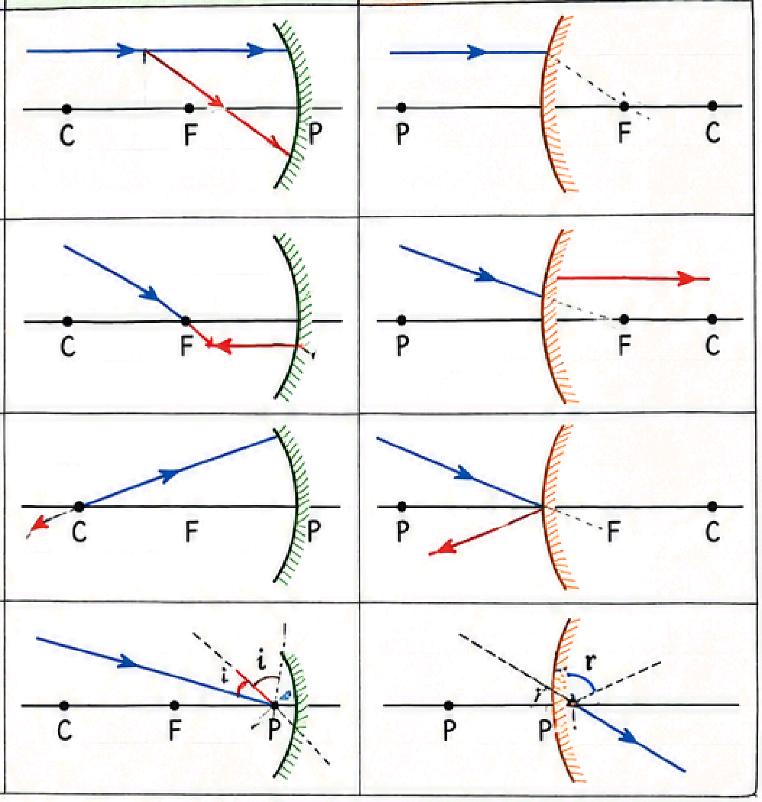
Object and image are equidistant from mirror.

PART 2: SPHERICAL MIRRORS

Ray Diagram Rules (4 standard rays)

1. **Parallel to axis**
   → after reflection passes through F (concave) or appears to come from F (convex)
2. **Through F**
   → after reflection goes parallel to axis
3. **Through C**
   → reflects back along same path
4. **At Pole (oblique)**
   → reflects making equal angles with principal axis

CONCAVE MIRROR | CONVEX MIRROR



For ray diagrams, use any <u>TWO</u> of the above. That's enough to find the image.

PART 3: REFRACTION OF LIGHT

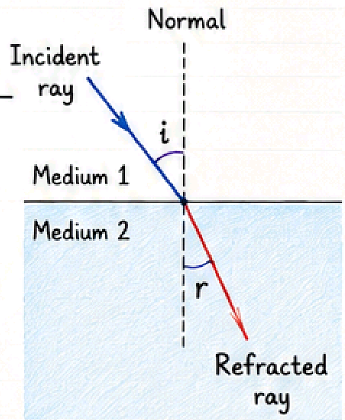
What is Refraction?

When light travels from one transparent medium to another, it changes its speed → this causes it to change direction (bend). This bending is called **Refraction.**

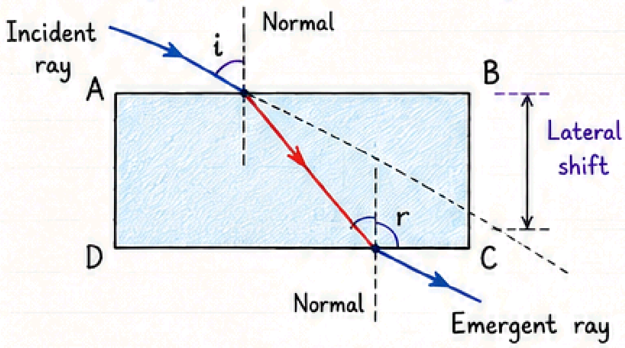
- **Rarer → Denser medium:** Light bends towards normal (slows down)
- **Denser → Rarer medium:** Light bends away from normal (speeds up)

Laws of Refraction

1. Incident ray, refracted ray, normal — all in the same plane
2. Snell's Law: $\frac{\sin i}{\sin r}$ = constant
   (for a given colour & pair of media)



Rectangular Glass Slab — Key Point



- Light bends at both surfaces (AB and CD)
- Emergent ray is parallel to incident ray
- But it's laterally shifted (displaced sideways)

This is because bending at both surfaces is equal and opposite, so net angular deviation = <u>0</u>.

PART 4: REFRACTIVE INDEX

Formula

$$n_{21} = \frac{\text{Speed in medium 1}}{\text{Speed in medium 2}} = \frac{v_1}{v_2}$$

Absolute Refractive Index:

$$n_m = \frac{c}{v} = \frac{\text{Speed in air/vacuum}}{\text{Speed in medium}}$$

Speed of light in vacuum = 3×10^8 m/s

Important Values to Remember

Medium	Refractive Index
Air	≈ 1.0003 (~1)
Water	1.33
Crown glass	1.52
Diamond	2.42 (highest common)
Ice	1.31

Optical density ≠ Mass density!

- Kerosene ($n = 1.44$) is optically denser than water ($n = 1.33$), but lighter in mass.
- Higher refractive index = Optically denser = Light travels slower.

PART 5: LENSES

Types of Lenses

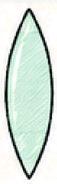
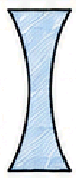
Convex Lens	Concave Lens
	
• Thicker at middle • Converging lens • f is positive	• Thicker at edges • Diverging lens • f is negative

Image Formation by Convex Lens

Object Position	Image Position	Size	Nature
At infinity	At F_2	Point-sized	Real, Inverted
Beyond $2F_1$	Between F_2 & $2F_2$	Diminished	Real, Inverted
At $2F_1$	At $2F_2$	Same size	Real, Inverted
Between F_1 & $2F_1$	Beyond $2F_2$	Enlarged	Real, Inverted
At F_1	At infinity	—	No image
Between F_1 & O	Same side as object	Enlarged	Virtual, Erect ☆

Pattern: Same as concave mirror!
Object between F and optical centre → Virtual image.

Image Formation by Concave Lens

Always gives:
Virtual + Erect + Diminished
— just like convex mirror!

Memory trick:
Concave Lens = Convex Mirror
(both always virtual, erect, diminished)

Lens Formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Don't mix up!

- Mirror formula: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$
- Lens formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Magnification (Lens)

$$m = \frac{h'}{h} = \frac{v}{u}$$

Note:
Lens magnification = v/u
(no negative sign unlike mirrors!)

PART 6: POWER OF A LENS

$$P = \frac{1}{f \text{ (in metres)}}$$

Unit: Dioptre (D)

- Convex lens → Positive power
- Concave lens → Negative power

For combined lenses: $P = P_1 + P_2 + P_3 + \ldots$ (just add powers!)

Example: Power = +2.0 D → $f = \frac{1}{2} = +0.5$ m (convex lens)

MOST COMMON EXAM MISTAKES

1. ✗ Using wrong sign for f in mirror formula
2. ✗ Confusing mirror formula (+ sign) with lens formula (− sign)
3. ✗ Forgetting that magnification sign tells real/virtual
4. ✗ Mixing up optical density with mass density
5. ✗ Using cm instead of metres for power calculation

QUICK REVISION TABLE

Concept	Mirror	Lens
Formula	$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$	$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
Magnification	$m = -\frac{v}{u}$	$m = \frac{v}{u}$
Concave f sign	Negative	Negative
Convex f sign	Positive	Positive
Always virtual	Convex mirror	Concave lens

IMPORTANT NUMERICAL SHORTCUTS

- ★ $R = 2f$ always
- ★ If $m = -2$ → real, inverted, 2× enlarged
- ★ If $m = +0.5$ → virtual, erect, half the size
- ★ Power (D) × focal length (m) = 1 always

How?

$P = \frac{1}{f} \rightarrow P \times f = 1$

Good luck! Focus on sign conventions — they're worth maximum marks in numericals!